CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
5th Semester of B. Pharm. Examination
University Theory Examination March-April 2018
PH317 PHARMACEUTICAL ANALYSIS-I

Date: 31.03.2018, Saturday **Time: 10 a.m. to 1.00 p.m.** **Maximum Marks: 80**

Instructions:

1. **There are three sections in this question paper.**
2. **SECTION – I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION – I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION – I is 30 minutes. Answers to SECTION – I must be written during the first 30 minutes of the examination.**
3. **Answers to SECTION – II and SECTION – III are to be provided in separate Main Answer Books provided to you.**
4. **Figures to right indicate marks.**
5. **Draw neat labeled sketches wherever necessary.**

SECTION - I

Q 1 Attempt all questions. Each question is of one mark. **20**

1. Solution of known concentration is known as__________.
[A] Titrant
[B] Titrand
[C] Indicator
[D] Primary standard

2. Following is a type of Aprotic solvent used in non-aqueous titration_________.
[A] CCl_4
[B] Benzene
[C] Hexane
[D] All of the above

3. Precipitation of one substance in a condition that will not allow precipitation of other substance which is called __________.
[A] Fractional precipitation
[B] Co-Precipitation
[C] No Precipitation
[D] Post Precipitation

4. Titration involving addition of excess of measured quantity of titrant to the sample solution and determination of unreacted titrant with another standard solution is called___________.
[A] Back titration
[B] Blank titration
[C] Direct titration
[D] Standardization

5. The greatest problem of inadequate sampling arises from________.

[A] Liquid
[B] Solid
[C] Gas
[D] All of above

6. Following is a type of titimetric analysis____________.

[A] Acid base neutralization titrations
[B] Redox titrations
[C] Complexometric titrations
[D] All of the above

7. Following is a type of redox titration___________.

[A] Potassium permanganate Titrations
[B] Potassium bromate Titration
[C] Ceric ammonium nitrate Titration
[D] All of the above

8. Which of the following is metallochromic indicator___________?

[A] Murexide
[B] Methyl orange
[C] $KMnO_4$
[D] Phenophthalein

9. RSD and CV can be calculated, if following values are given__________.

[A] Mean
[B] Standard deviation
[C] Standard deviation and Median
[D] Standard deviation and Mean

10. According to Arrhenius theory acid is a conpound that___________.

[A] Ionizes in water to give proton
[B] Ionizes in water to give hydroxyl ion
[C] Has a tendency to lose a proton.
[D] Accept an electron pair

11. Following is a salt derived from weak acid and strong base__________.

[A] Sodium chloride
[B] Ammonium chloride
[C] Sodium acetate
[D] Ammonium acetate

12. An indicator used in Fajan's method is__________.

[A] Ferric Alum
[B] Chromate salt
[C] Fluorescein
[D] Methyl orange

13. Karl Fischer titration is use to determine__________.

[A] % purity of drug
[B] Water content
[C] Heavy metal impurities
[D] All of above

14. A solution containing equal amount of acid and its salt has ___________ buffer capacity.

[A] Moderate
[B] Maximum
[C] Minimum
[D] One half

15. In Volhard's method there is ____________.

[A] No precipitation
[B] Utilization of adsorption indicator
[C] Formation of soluble colored product.
[D] Formation of colored precipitates

16. Following titrant is served as its own indicator__________.

[A] Sodium thiosulphate
[B] Sodium hydroxide
[C] Iodine
[D] Potassium permanganate

17. What is added in complexometric titration for stability of complexes formed?____________

[A] Salt
[B] Antioxidants
[C] Preservatives
[D] Buffer solution

18. If molar concentration of ions greater than Ksp___________.

[A] No precipitation
[B] Precipitation
[C] Unsaturated solution
[D] Rapid reaction

19. Which of the following statistical test is used to compare results of two different analytical methods?

[A] Q-test
[B] F-test
[C] t-test
[D] None of the above

20. If the number of moles of solute is dissolved in one liter solvent is called__________.

[A] Normality
[B] Molarity
[C] Molality
[D] % Weight/volume

SECTION – II

Q 2	**Attempt any TWO of the following**		
A	Write a note on acid-base indicators.	**05**
B	Describe in detail common ion effect with suitable example.	**05**
C	What is complexometric titration? Explain general principle of complexometric titration and give its application.	**05**
Q 3	**Attempt any TWO of the following**	
A	What is gravimetric analysis? Describe in detail coprecipitation.	**05**
B	Write a detail note on Mohr's precipitation method.	**05**
C	Give detail account on leveling and differentiating solvent.	**05**

SECTION – III

Q 4	**Attempt any FOUR of the following**	
A	What is Indian Pharmacopoeia? Give brief overview on general contents of monograph.	**05**
B	Enlist various types of redox titration. Differentiate between iodimetric and iodometric titration.	**05**
C	Explain in brief neutralization curve (titration cure) for acid-base titration of acetic acid and sodium hydroxide.	**05**
D	Write a note on different types of pharmacopoeial water and enlist various QC tests for waters.	**05**
E	Give comparative overview on effect of masking and demasking in complexometry.	**05**
F	Exaplin in detail principle of Karl-Fisher titration and give its applications.	**05**
Q 5	**Attempt any FOUR of the following**	
A	Write a note on sampling and sampling techniqes.	**05**
B	What are buffers? Expalin in brief Henderson Hesselbach equation and applications of Buffers in pharmacy.	**05**
C	Describe different types of error and its minimization.	**05**
D	Give applications of aqueous and non-aqueous acid-base titrations.	**05**
E	Describe in brief about Redox titration and its applications. Give brief account on Redox indicators.	**05**
F	What is primary standard? Give ideal requirements of primary standards.	**05**

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